Ossp skill(week -2)

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1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    while (1)
   {
        // Display shell prompt
        printf("myshell$ ");
        fflush(stdout);
```

```c
// Read user input
if (fgets(input, sizeof(input), stdin) == NULL)
{
    printf("\n");
    break;
}

// Remove newline
input[strcspn(input, "\n")] = '\0';

// Ignore empty input
if (strlen(input) == 0)
    continue;

// Handle exit command
if (strcmp(input, "exit") == 0)
{
    printf("Exiting shell...\n");
    break;
}

// Split input into arguments
int argc = 0;
char *token = strtok(input, " ");

while (token != NULL && argc < MAX_ARGS - 1)
{
    args[argc++] = token;
    token = strtok(NULL, " ");
}

args[argc] = NULL;
```

```c
// Built-in cd command
if (strcmp(args[0], "cd") == 0)
{
    if (args[1] == NULL)
    {
        fprintf(stderr, "cd: missing directory\n");
    }
    else
    {
        if (chdir(args[1]) != 0)
            perror("cd");
    }

    continue;
}

// Create child process
pid_t pid = fork();

if (pid < 0)
{
    perror("fork");
    continue;
}

// Child process
if (pid == 0)
{
    execvp(args[0], args);

    // If execvp fails
```

```
            perror("Command failed");
            exit(EXIT_FAILURE);
        }

        // Parent process
        else
        {
            waitpid(pid, NULL, 0);
        }
    }

    return 0;
}
```

Output:

```
lenovo@SAISHASHANK:~$ nano skill2.c
lenovo@SAISHASHANK:~$ gcc skill2.c -o skill2
lenovo@SAISHASHANK:~$ ./skill2
myshell$ ls
a.out                 producerconsumer                   program5.c
command_executor      producerconsumer.c                 program5.c:Z
command_executor.c    program1.c                         program5.txt
exp1                  program1.c:Zone.Identifier         program5.txt
exp1.c                program1.txt                       program6.c
exp1.c.save           program1.txt:Zone.Identifier       program6.c:Z
exp2.c.save           program2.c                         program6.txt
exp2.c.save.1         program2.c:Zone.Identifier         program6.txt
exp2.c.save.2         program2.txt                       program7.c
experiment            program2.txt:Zone.Identifier       program7.c:Z
experiment.c          program3.c                         program7.txt
filename              program3.c:Zone.Identifier         program7.txt
fork_exec             program3.txt                       program8.c
fork_exec.c           program3.txt:Zone.Identifier       program8.c:Z
hello.c               program4.c                         program8.txt
input.txt             program4.c:Zone.Identifier         program8.txt
ossp                  program4.txt                       program9.c
output.txt            program4.txt:Zone.Identifier       program9.c:Z
myshell$ date
Tue Aug 25 03:01:48 UTC 2026
myshell$ pwd
/home/lenovo
myshell$ whoami
lenovo
myshell$ cd
cd: missing directory
myshell$
```